Application of Advanced Deep Learning Techniques for Face Detection and Age Estimation

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***Abstract*—In recent years, AI-based algorithms have been used to solve a wide range of previously intractable issues. Face detection and recognition from large groups, as well as feature identification from a description, are two of the most important security issues that can be addressed with image processing and modern AI. In this research, we use deep learning to analyze facial images to detect faces and estimate ages. The proposed method was used on photos of King Abdullah bin Abdulaziz AL Saud to determine his face and approximate his age. For this purpose, we relied on powerful deep learning models like DELWO and MTCNN to perform tasks like identifying faces and determining how old they are. The findings are very encouraging and promising. The experimental results have shown that the proposed method has good accuracy for face recognition (98.50%) and age estimation (82%).**

Keywords*—*face recognition, age prediction, deep learning, convolutional neural network, King Abdullah bin Abdulaziz.

I. Introduction

King Abdullah bin Abdulaziz AL Saud biography documentation center has a large dataset that contain a large set of King Abdullah bin Abdulaziz images, and they do not have enough information about the image like history, age, and era. The Centre aims to know and record the age or era of each image manually and this is very hard to do. According to that, the main problem that we claim to solve is to find the age of King Abdullah in each image automatically using deep learning approach. This also includes detection and recognition of King Abdullah face in those images.

In this paper, we aim to utilize computer capabilities to solve this problem for several reasons. First, it will preserve the history of King Abdullah bin Abdulaziz and organize the available dataset. Second, this image-based dataset is utilized for the deep learning model training and tuning to automatic age estimation of the King Abdullah.

Generally, the classical machine learning and other statistical models requires feature engineering to extract manually relevant features. Deep learning models, on the other hand, have the capability to extract automatically the various semantic features from the image or video-based dataset, which no other machine learning model can do. we realized that for estimating the age from the King Abdullah bin Abdulaziz archives image, first we need to understand how to detect his face form the image dataset. By studying the various literatures on the face detection, we get the detailed idea about how the image processing, face recognition and age detection work with the use of deep learning algorithms. Using various pre-trained deep learning models in pair with the computer vision tools, the face can be detected using the pre-trained models and the face detection accuracy will extend at certain limits. Once the face is detected from the image dataset, the next step of the algorithm consists in checking whether this face is belonging to King Abdullah bin Abdulaziz or not. If the face is belonging to King Abdullah bin Abdulaziz, then the algorithm will estimate the age of King Abdullah bin Abdulaziz. For the age estimation, NCNN model is used which is trained with the King's image dataset and tuned for better accuracy.

II. Related Work

In this section, we present the previous research studies related to face detection, facial recognition methodologies, and age estimation.

Face detection has been intensively studied in the past decades because of its wide applications in face analysis. As an important processing step for face recognition, a robust detection algorithm is expected to identify faces under arbitrary image conditions.

The proposed methods in [1, 2] are traditional methods for face detection. In [1], Viola and Jones proposed the first algorithm that made face detection practically feasible in real-world applications and widely used in digital cameras and photo organization software. Since then, research in face detection has made significant progress in the direction of providing algorithms that are able to detect faces 'in-the-wild. In [2], joint Haar-like features were proposed, which are based on the co-occurrence of multiple Haar-like features. It was claimed that feature co-occurrence can better capture the characteristics of human faces, making it possible to construct a more powerful classifier.

Other studies were based on deep learning for face detection. In [3] a Cascade CNN was originally designed to address the issue of high computing cost and significant variances. In [4], An ICC-CNN was proposed to reject samples in different levels of a single CNN. One of the advantages of these strategies is their quick calculating speed.

For face recognition, in [5] Holistic matching methods are introduced using the Eigenface approach. In this method the facial region mask is took as an input source dataset for catching the face from the entire image to create an Eigenface.

In [6], a feature-based method was proposed (facial structure-based). This algorithm is based on the locations of facial features such as the mouth, nose, and eye which form the facial image data.

these conventional methods of facial recognition were too complex and after collecting the features from the dataset the selection process of the classification model is also a very haptic task. Fortunately, the improvement in the graphic card hardware (GPU) makes possible real-time face recognition. This also allows to create a more hyper-parameter model generation. The deep learning model can extract automatically the features from the image-based dataset [7, 8]. However, this kind of methods requires large amount of dataset to get good performance for the face recognition. In [9], facial images are recognized using the CNN model which was developed using the Python and Keras deep learning frameworks [10]. In [11], the authors also suggested the use of Let-Net-5 CNN model configuration for face detection. In [12], the authors proposed ArcFace which is a machine learning model that takes two face images as input and outputs the distance between them to see how likely they are to be the same person. This can be used for face recognition and face search.

For age estimation, the problem can be formulated as a classification problem in which the person age classified into various groups, and check in which age group this person is belongs to. It can be also formulated as a regression-based problem [13, 14]. In [15, 16], the age estimation is based on the classification approach. In [17], an ordinal classification estimation for the age detection problem was proposed. In these algorithms the hyperplane of the estimation models split the facial image information into two categories the relative order of the image and age which are available in the model dataset during the aggregation. This same process was also utilized by [18] in their research work. In [19], the age classification is based on hierarchical algorithm. This algorithm is able to utilize the global as well as the local features from the facial image data information [20].

In [21,22], the proposed age estimation model was based on the regression-based approach. The proposed model uses a CNN and an SVR. In [23], the authors used the CRCNN based framework which is a novel approach for the age estimation problem. In [24], the authors used ResNet50 as transfer learning model of CNN architectures [25]

The study in [24] Propose an age estimation from the facial data from the convolution neural network model from the large image dataset and process the images by the computer vision algorithms. This research works also claims that by using the computer vision it required less data for model training as compared to their models. For the model training they used the ResNet50 configuration of CNN model which has a high dimension so training data reverent is apparently low. Further the model performance compared using the mean absolute error algorithm. This ResNet50 is also known as transfer learning model of CNN algorithm [25].

III. The Proposed Solution

In this section, we focus on data collection and preparation, face detection, face recognition, and age estimation (see Fig. 1) using advanced deep learning techniques such as DELWO [26].

A. Data Collection

The first step is related to data collection. We collected the image data set from King Abdullah bin Abdulaziz Al Saud biography documentation center which has collected an archive of the history of King Abdullah bin Abdulaziz Al Saud. This involves the various life events from his young age until his death. The number of images is 21000 images. We applied face detection to these images and extracted the faces from them. We gathered 3561 images as faces.

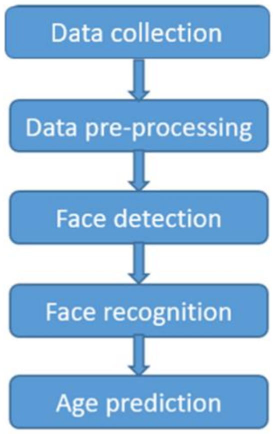

Fig. 1. The main steps of the proposed system..

This involved many public events, public speeches, celebration type of events. Few samples of the dataset are depicted in Fig 2.

Fig. 2. Samples data from the King Abdullah bin Abdulaziz Al Saud biography documentation center.

B. Data Pre-processing

We used data augmentation technique to increase the efficiency of CNN and minimize overfitting issues. Data augmentation involves changing randomly selected training pictures. The idea is to provide additional training data while exposing the training model to various distributions of real-world properties such as rotation, shear, width shift, and height shift as well vertical and horizontal flips (see Fig. 3).

Fig. 3. Data Augmentation.

C. Face Detection

For face detection we used different algorithms. We compared between MediaPipe and MTCNN. MediaPipe Face Detection is an ultrafast face detection solution that comes with 6 landmarks and multi-face support. It is based on BlazeFace [27], a lightweight and well-performing face detector tailored for mobile GPU inference. MTCNN [28] which stands for Multi-task Cascaded Convolutional Network is a modern tool for face detection, leveraging a 3-stage neural network detector.

We tested MediaPipe and MTCNN using 120 images. MediaPipe succeeded with 80% while MTCNN succeeded with 93%. That is why we selected MTCNN algorithm for face detection.

D. Face Recognition

For face recognition, we used the DELWO system introduced in [26] and shown in Fig 4. DELWO is a generic model in the sense it is possible to use and combine different pretrained deep learning models such as VGG16, VGG19, ResNet50, AlexNet, and so forth.

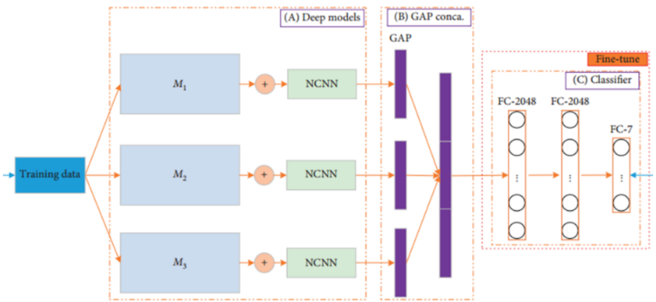

Fig. 4. The DELWO system [26].

The dataset is split into 80% for training, 12% for testing, and 8% for validation. The dataset is split into two classes. Class 1 represents the face images of King Abdullah, and class 2 represents the face images of other persons.

The whole framework of our proposed algorithm consists of four blocks. Block one represents the original input image, block two normalizes the input image, and block three represents the NCNN (Nonsequential CNN) model to carry out the NCNN learning. Block four explains the multiple models that combine many models to study more parts of the "truth directly" to take advantage of ensemble learning.

In the training phase, the training data is fed into three deep learning models (VGG16, ResNet50, and VGG19), after which the GAP layers are merged into one long layer. NCNN module consists of branches; each branch has different filters with different sizes. Finally, the classifier module receives the combined information from the three deep models and does weight optimization.

The testing data is fed into the trained model to calculate the final prediction probability for the testing phase.

The benefit of combining an NCNN module with a GAP layer is that the number of trainable parameters can be reduced significantly. As a result, our model will be lighter, making the mitigating model overfitting and promoting generalization ability.

E. Age Prediction

In the age prediction, the purpose of this is to predict the date of King Abdullah images. The steps of the detection are depicted on Fig 5. We have three main steps:

Step 1:

We build a model using DELWO to predict whether the King face is before 2000 or after 2000. In this model we train the model using two classes, one class contains all face images of King Abdullah before 2000 and the other class contains all face images after 2000.

Step 2:

If the King face in step1 is predicted as before 2000, we use another model using DELWO to predict the period of this face within six classes, where each class contains the King face in (1930-1950, 1950-1960, 1960-1970, 1970-1980, 1980-1990, 1990-2000)

Step 3:

If the King face in step1 is predicted as after 2000, we use a DELWO model to predict the year of King face between 2000 and 2015. The model is trained with 13 classes (2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008-2009, 2010, 2011, 2012, 2013-2015).

The models used for age estimation are almost the same as the one used for face recognition. We do some changes in the hyperparameters.

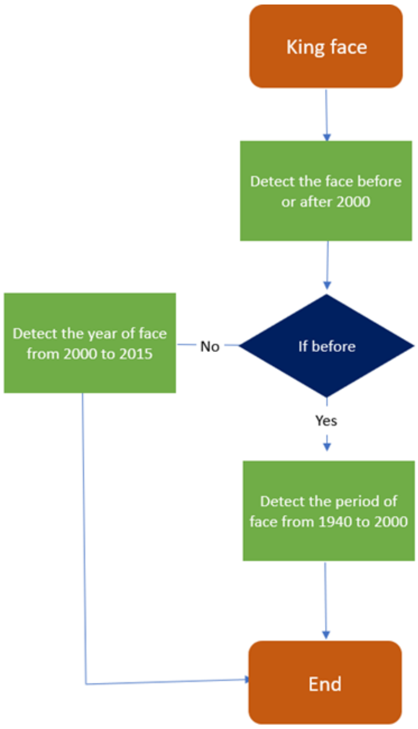

Fig. 5. Age estimation using DELWO models.

IV. DATA ANALYSIS AND RESULTS

This section is based on the King's image data analysis and various processing performed on the programming environment. After that, we explain how the deep learning model is implemented in the programming environments and how model training and testing are carried out.

A. *Image Data Analysis*

For the King's image-based dataset analysis, we used a Python environment. Python provides many packages for image data processing, machine learning and deep learning.

the input images have different size. The deep learning model requires that all the images should have the same size and dimensions. All the images were resized into (224*224 pixel) images to fit the deep learning model input. Once the image data are resized, the images are converted into the Color image format (RGB).

Now all the face images are processed, and the ROI process is completed, the next step is to split the dataset into training and testing datasets.

B. *Model Implementation Steps and Results*

Once the image of the King is recognized then the next step is to predict the age of the King in various range.

The model structure of the Deep CNN model (DELWO) was described in the previous section. The different steps of the whole process are shown on Fig 6.

Fig. 6. Face detection, face recognition, and age prediction steps.

For the model training, the data was split into 80% for training, 12% for testing, and 8% for validation. For face recognition, the obtained validation accuracy was 99.6% while the obtained testing accuracy was 98.5%. The results are depicted on Fig 7.

For the three models of age prediction, the model in step 1 got a validation accuracy of 85% and a testing accuracy of 85% as shown in the Fig 8. The model in step 2 got a validation accuracy of 75% and the testing accuracy was 80% as shown in the Fig 9. The model in step 3 got the validation accuracy of 87% and the testing accuracy was 82% as shown in Fig 10.

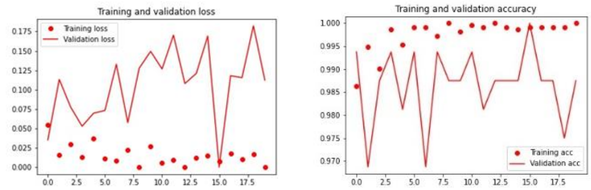

Fig. 7. Training, validation accuracy and loss for face recognition.

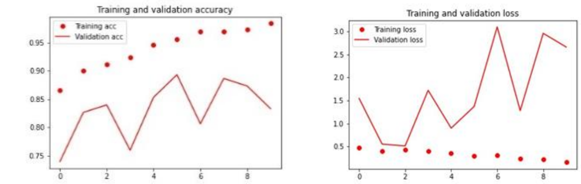

Fig. 8. Training, validation accuracy, and loss for step 1 in age prediction.

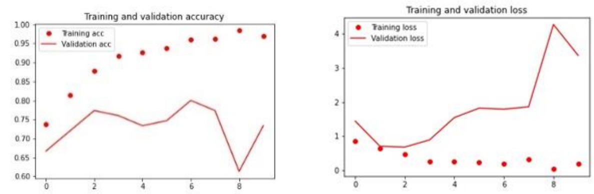

Fig. 9. Training, validation accuracy, and loss for step 2 in age prediction.

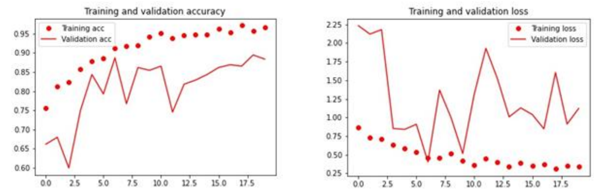

Fig. 10. Training, validation accuracy, and loss for step 3 in age prediction.

We compared our results for face recognition with other models in [26] as shown in Table 1. In fact, we obtained very good results for the face recognition step.

The obtained results are promising and compete with existing works in the literature [17]. We compared the accuracy of the age estimation of all 8 classes with vgg16, vgg19, and resnet50 as shown in table 2. The accuracy will increase if we reduce the number of classes. Our dataset is very difficult because we have a lot of images with bad quality so the accuracy in the age estimation is not very high, but it competes and even outperforms other existing methods.

TABLE I. COMPARISON OF THE PROPOSED APPROACH WITH EXISTING TECHNIQUES FOR FACE RECOGNITION

Model	Accuracy
VGG16_Model	94%
VGG16_NCNN	96%
VGG19	95%
VGG19_NCNN	96.5%
ResNet50	97%
ResNet50_NCNN	97.5%
DELWO	**98.5%**

TABLE II. COMPARISON OF THE PROPOSED APPROACH WITH EXISTING TECHNIQUES FOR AGE ESTIMATION

Model	Accuracy
VGG16_Model	64%
VGG16_NCNN	66.5%
VGG19	65%
VGG19_NCNN	64%
ResNet50	67%
ResNet50_NCNN	69%
DELWO	82%

The results obtained using a sample of images in different periods of King Abdullah life are shown on Fig 11.

Fig. 11. The obtained results using a sample of King Abdullah images.

V. CONCLUSION AND FUTURE WORK

In this paper, we presented a system for face detection, face recognition, and age estimation. The proposed approach is based on advanced deep learning techniques such as MTCNN and DELWO architectures. We applied the techniques using King Abdullah images obtained from the documentation center. The obtained results are very encouraging and very promising. As future work, we expect using data of better quality to improve the accuracy of the approach. The proposed approach can be extended in different ways and can be used for other renowned personalities.

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